Epiphyte Biodiversity & Tree Size Field Study

Research question: Does tree size influence epiphyte species richness on pin oak trees at UBC?

Project Overview

Investigated the relationship between tree size and epiphyte biodiversity on pin oak trees at UBC. Field observations were used to examine whether larger trees supported greater epiphyte species richness and to assess patterns in epiphyte abundance across the sampled community.

Field Methods

- Sampled epiphytes on 3 pin oak trees using quadrants positioned at multiple heights and orientations around each trunk
- Identified bryophyte and lichen species and recorded their abundance within each sampling quadrat.
- Measured tree circumference and converted measurements to diameter at breast height (DBH) for comparison with species richness.
- Combined field observations with the class dataset for analysis across 27 trees.

Data Analysis

- Compared tree DBH with epiphyte species richness using correlation and regression statistics.
- Created a scatter plot to visualize the relationship between tree size and species richness.
- Developed a rank-abundance curve to evaluate the relative abundance and evenness of epiphyte species across the sampled community.

Results

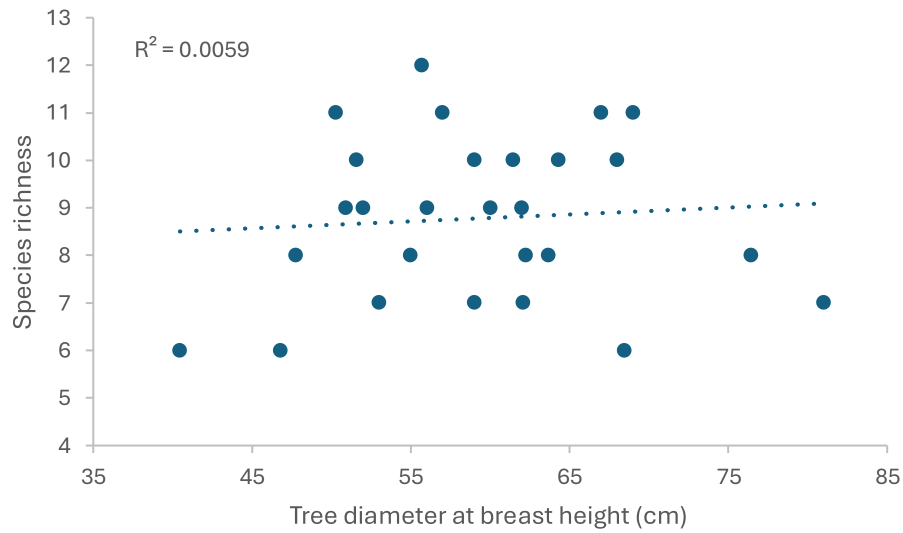

Figure 1. Species richness of epiphytes has no relationship observed with the diameter of trees at breast height (cm) (R^2=0.0059,r=0.077,N=27). Data was collected from pin oak trees at ubc on June 5th 2026

Result: Tree diameter showed little association with epiphyte species richness. The two-tailed p-value of 0.70 provided little evidence of a statistically meaningful relationship in this dataset.

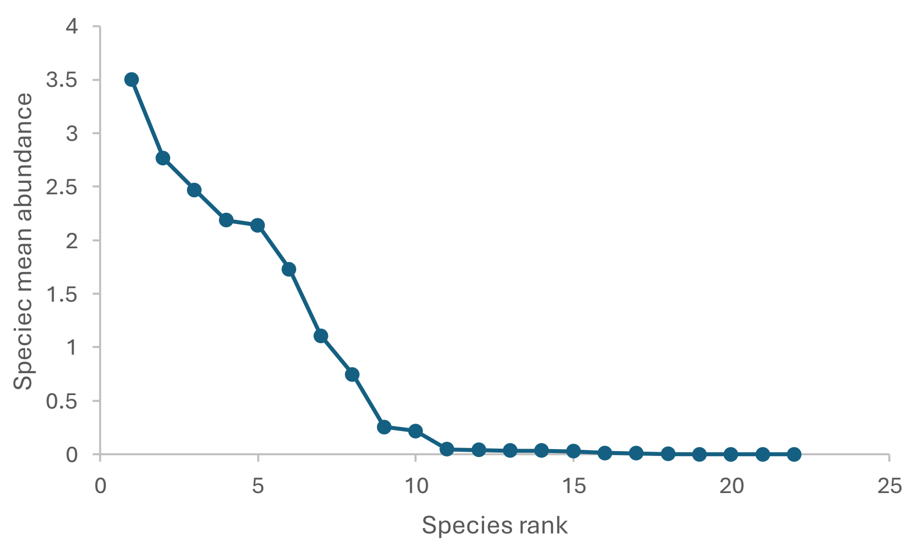

Figure 2. Rank abundance curve of mean number of epiphyte species. Each point represents average number of epiphyte species across all trees (N=27). Data was collected from pin oak trees at ubc on June 5th 2026.

Results: The rank-abundance curve indicates an uneven epiphyte community. A small number of species occurred at relatively high abundance, while many species were present at much lower abundance.

Interpretation

Larger trees could theoretically support greater epiphyte richness because increased surface area may increase the probability of encountering additional species, while greater habitat variety may create more microhabitats with different moisture and light conditions. Larger habitat areas may also support larger populations and reduce local extinction risk.

Observed Pattern: Despite these expectations, tree size explained very little variation in epiphyte species richness in the sampled dataset. Species richness remained variable across both smaller and larger trees.

Conclusion

Tree diameter was not a strong predictor of epiphyte species richness in the sampled pin oak trees at UBC. Although larger trees may theoretically provide greater surface area and habitat diversity, the observed data suggest that tree size alone did not explain variation in epiphyte richness. The rank-abundance analysis also indicated an uneven community in which a small number of species were substantially more abundant than others.

Limitations

Tree diameter was only one potential factor affecting epiphyte biodiversity. Future studies could separately examine environmental variables such as moisture, light exposure, bark characteristics, and microhabitat conditions to better understand variation among trees.